Tick ✓ the correct option in the following :

A force acting on a particle is said to be _____ if its line of action always passes through a fixed point.

(a) applied force (b) friction (c) central force (d) none of these

The central force is _____ if it is directed towards its centre.

(a) attractive (b) not attractive (c) repulsive (d) none of these

The central force is _____ if it is directed away from the centre.

(a) attractive (b) repulsive (c) not repulsive (d) none of these

Orbit of a particle moving under a central force is necessarily _____.

(a) a space curve (b) a plane curve (c) a straight line (d) none of these

The orbit described under a central attractive force varying directly as the distance is _____ having the centre at the centre of the force.

(a) an ellipse (b) a circle (c) a straight line (d) none of these

A central force $\vec{F}$ is said to be conservative force if $\nabla \times \vec{F} =$

(a) 3 (b) 2 (c) 1 (d) 0

In a conservative field, the total energy of a particle _____ throughout the motion.

(a) varies rapidly (b) does not remain constant

(c) remains constant (d) none of these

A central force is conservative if and only if it is a function of

(a) velocity and acceleration (b) distance and velocity

(c) distance alone (d) none of these

Areal speed of the radius vector $\vec{r}$ joining the particle to the centre of force is

(a) not constant (b) constant

(c) variable (d) none of these

The component of force along the radius vector is

(a) $F_r = m\left[\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right]$ (b) $F_\theta = \frac{m}{r}\frac{d}{dt}\left(r^2\frac{d\theta}{dt}\right)$

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(c) $F_r = m\left[\frac{dr}{dt} - r\left(\frac{d\theta}{dt}\right)\right]$ (d) none of these

(xi) The component of force perpendicular to the radius vector is

(a) $F_r = m\left[\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right]$ (b) $F_\theta = \frac{m}{r}\frac{d}{dt}\left(r^2\frac{d\theta}{dt}\right)$

(c) $F_r = m\left[\frac{dr}{dt} - r\left(\frac{d\theta}{dt}\right)\right]$ (d) none of these

(xii) When the force $\vec{F}$ is an attractive force directed towards origin, then $F_r = -F$ and $F_\theta =$

(a) 3 (b) 2 (c) 1 (d) 0

(xiii) The differential equation of the orbit in polar coordinates is $\frac{d^2u}{d\theta^2} + u =$

(a) $\frac{f}{hu}$ (b) $\frac{f}{h^2u^2}$ (c) $\frac{f}{h^3u^3}$ (d) none of these

(xiv) The length of the radius vector at an apse is known as the apsidal

(a) acceleration (b) velocity (c) distance (d) none of these

(xv) The line joining an apse to the centre of force is _____ line.

(a) a focal line (b) an apse (c) an initial (d) none of these

(xvi) The angle between two consecutive apse lines is an

(a) angle of friction (b) angle of projection

(c) apsidal angle (d) angle of depression

(xvii) The analytic condition for a point with radius vector r to be an apse is $\frac{dr}{d\theta} =$

(a) 1 (b) 0 (c) -1 (d) -2

(xviii) If ϕ is the angle between the radius vector and the tangent at some point, then $\tan\phi =$

(a) $r\frac{d\theta}{dr}$ (b) $\frac{1}{r}\frac{dr}{d\theta}$ (c) $\frac{dr}{d\theta}$ (d) none of these

(xix) If ϕ is the angle between the radius vector and the tangent at some poin then $\cot\phi =$

(a) $r\frac{d\theta}{dr}$ (b) $\frac{1}{r}\frac{dr}{d\theta}$ (c) $\frac{dr}{d\theta}$ (d) none of these

(xx) Kepler's _____ Law states that each planet describes an ellipse wi the sun at one focus.

(a) Third (b) Second (c) First (d) none of thes

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(xxi) Kepler's _____ Law states that the radius vector drawn from the sun to the planet sweeps out equal areas in equal times.

(a) Third (b) Second (c) First (d) none of these

(xxii) Kepler's _____ Law states that the squares of the periods of the planets are proportional to the cubes of the semi-major axes of their orbits.

(a) Third (b) Second (c) First (d) none of these

(xxiii) The force which is always directed towards or away from a fixed point is called:

(a) Conservative force (b) Central force

(c) Internal force (d) Conservative force

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(xxiv) The necessary and sufficient condition for a particle to be in equilibrium at a point is that _____ at that point.

(a) $\nabla V = 1$ (b) $\nabla V = 2$ (c) $\nabla V = 0$ (d) $\nabla V \neq 0$

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